

Early Detection of Polycystic Ovary Syndrome Using All of Us Data

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All of Us
RESEARCH PROGRAM

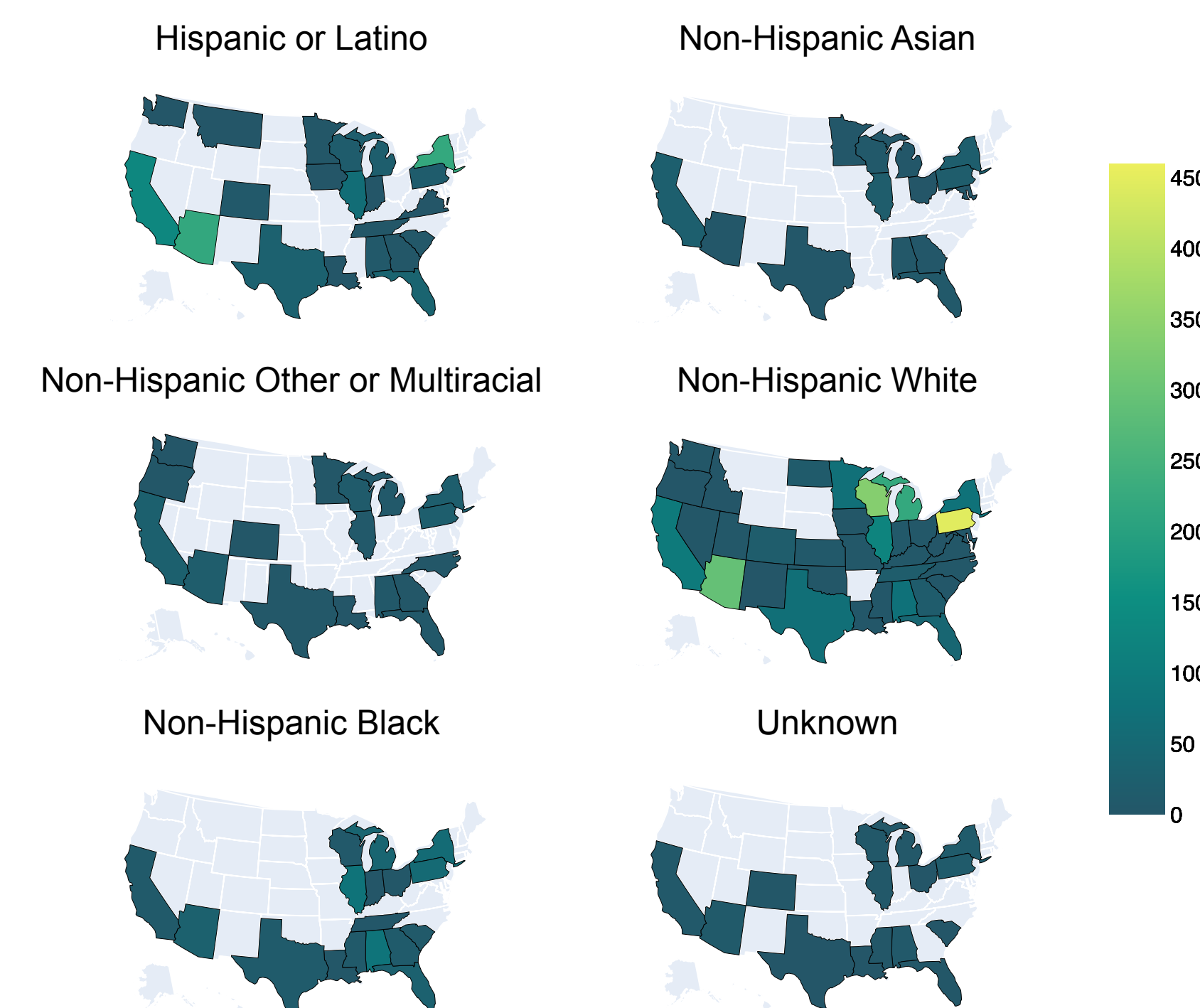
Introduction

- Polycystic Ovary Syndrome (PCOS) is one of the most common endocrine conditions, affecting ~11-13% of reproductive-aged women globally.¹
- Despite its high prevalence, PCOS remains under recognized and understudied, with diagnosis often requiring multiple clinical visits and extended timeline.¹
- PCOS carries broad and lifelong health risks, including: increase risks of type 2 diabetes, cardiovascular disease, infertility, pregnancy complications and endometrial cancer.¹ Delayed or missed diagnosis may limit the opportunities for early management and preventions of these complications
- Evidence from a large international survey shows that over a third of women experienced delays >2 years and consulted >3 health professionals before diagnosis.²
- Early detection of PCOS remains an important yet understudied challenge, particularly across racially and socioeconomically diverse populations. Large-scale, nationally representative datasets such as the All of Us Database complemented with Area Deprivation Index (ADI) scores³ provide new opportunities to investigate early risk signals at the population level.

1. Stener-Victorin E, Teede H, Norman RJ, et al. Polycystic ovary syndrome. Nat Rev Dis Primer. 2024;10(1):27. doi:10.1038/s41572-024-00511-3
2. Gibson-Helm M, Teede H, Dunaff A, Dokras A. Delayed Diagnosis and a Lack of Information Associated With Dissatisfaction in Women With Polycystic Ovary Syndrome. J Clin Endocrinol Metab. 2017;102(2):604-612. doi:10.1210/clinem.2016-2963
3. Kind AJH, Buckingham WR. Making Neighborhood-Disadvantage Metrics Accessible — The Neighborhood Atlas. N Engl J Med. 2018;378(26):2456-2458. doi:10.1056/NEJMp1802313

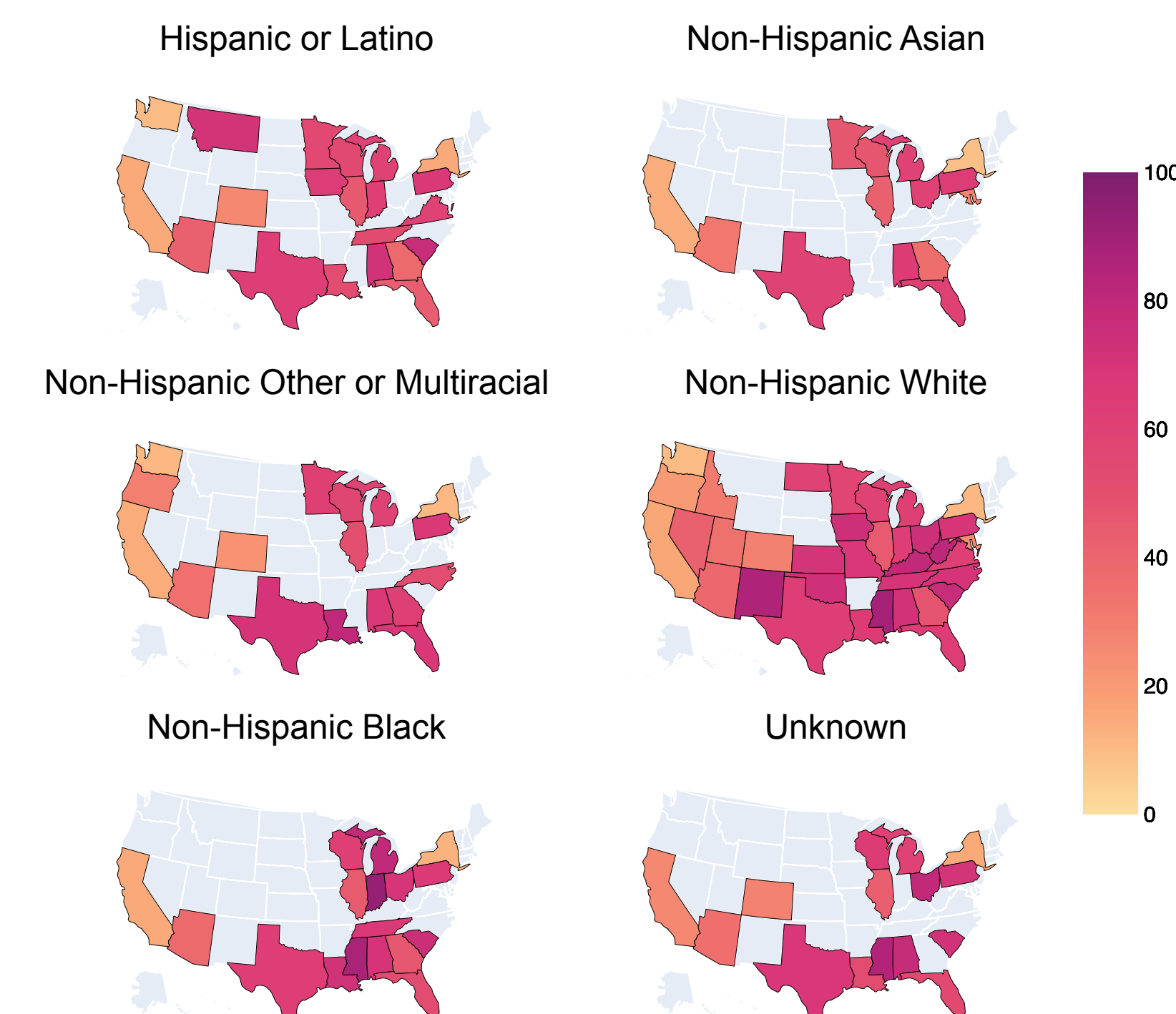
PCOS Patient Demographic Distribution (All of Us)

Count of PCOS individuals by State per Race/Ethnicity



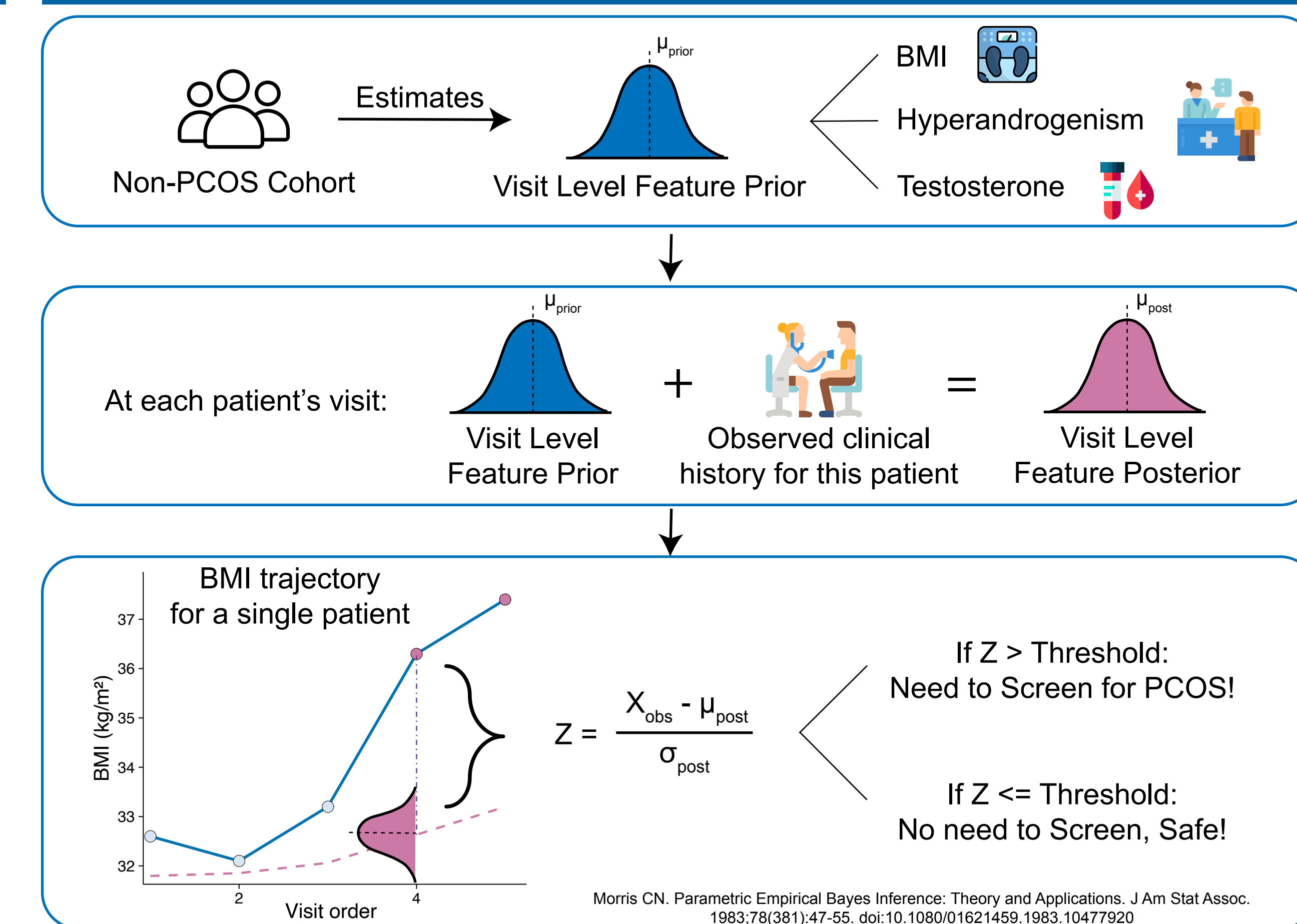
• 4166 total patients meeting PCOS criteria

Aggregated ADI of PCOS individuals by State per Race/Ethnicity

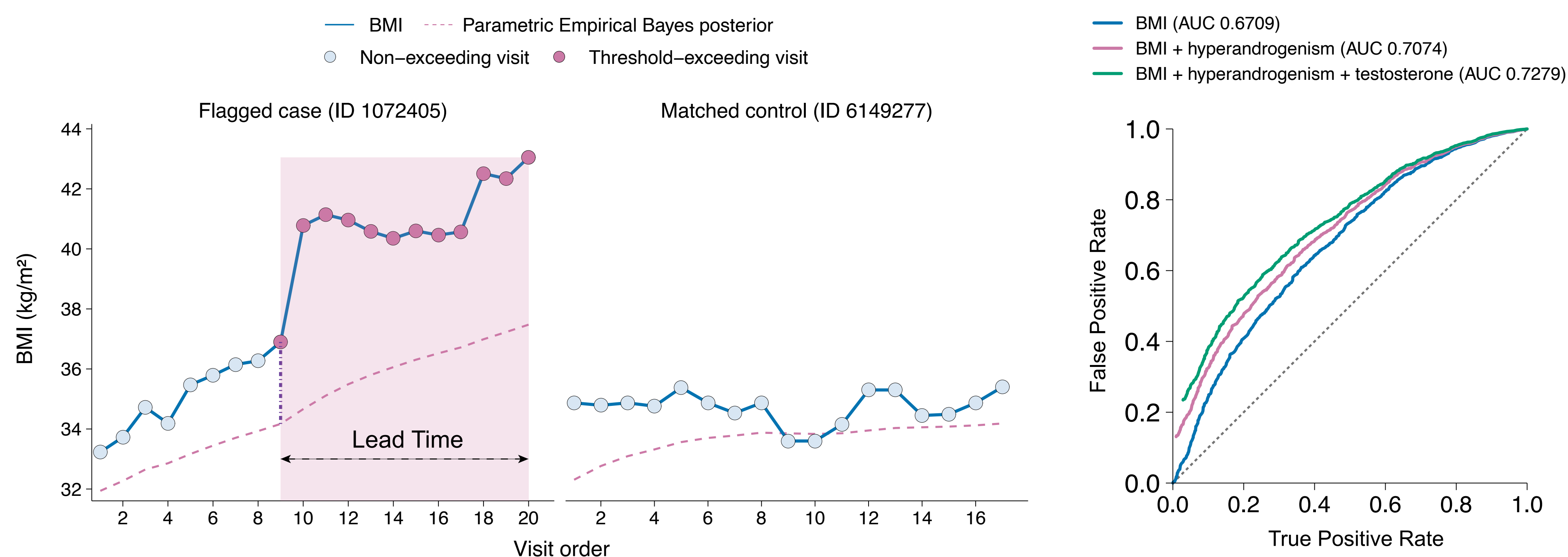


- 0: low levels of adverse social exposome,
- 100: high levels of adverse social exposome³

Methods



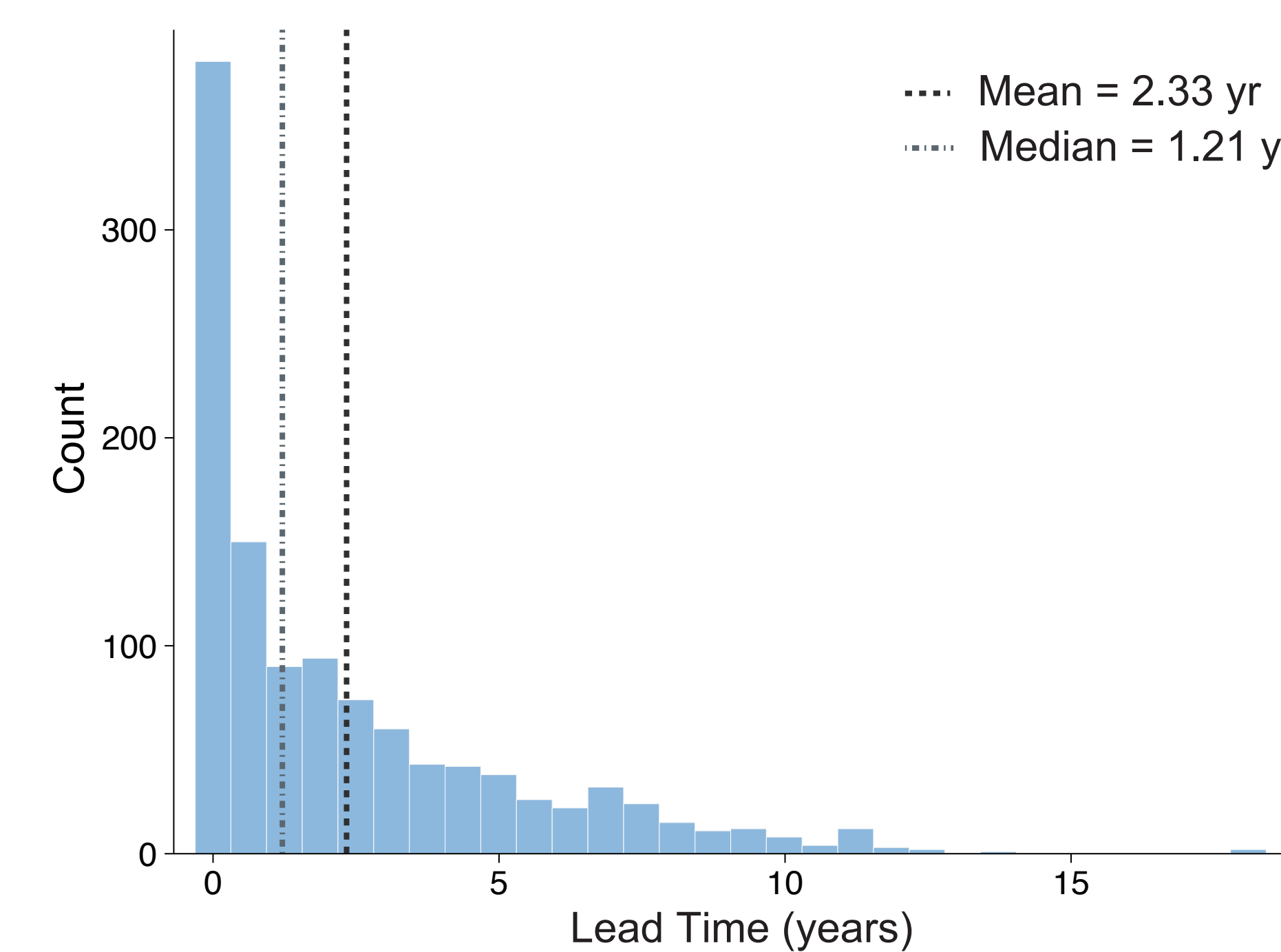
Results



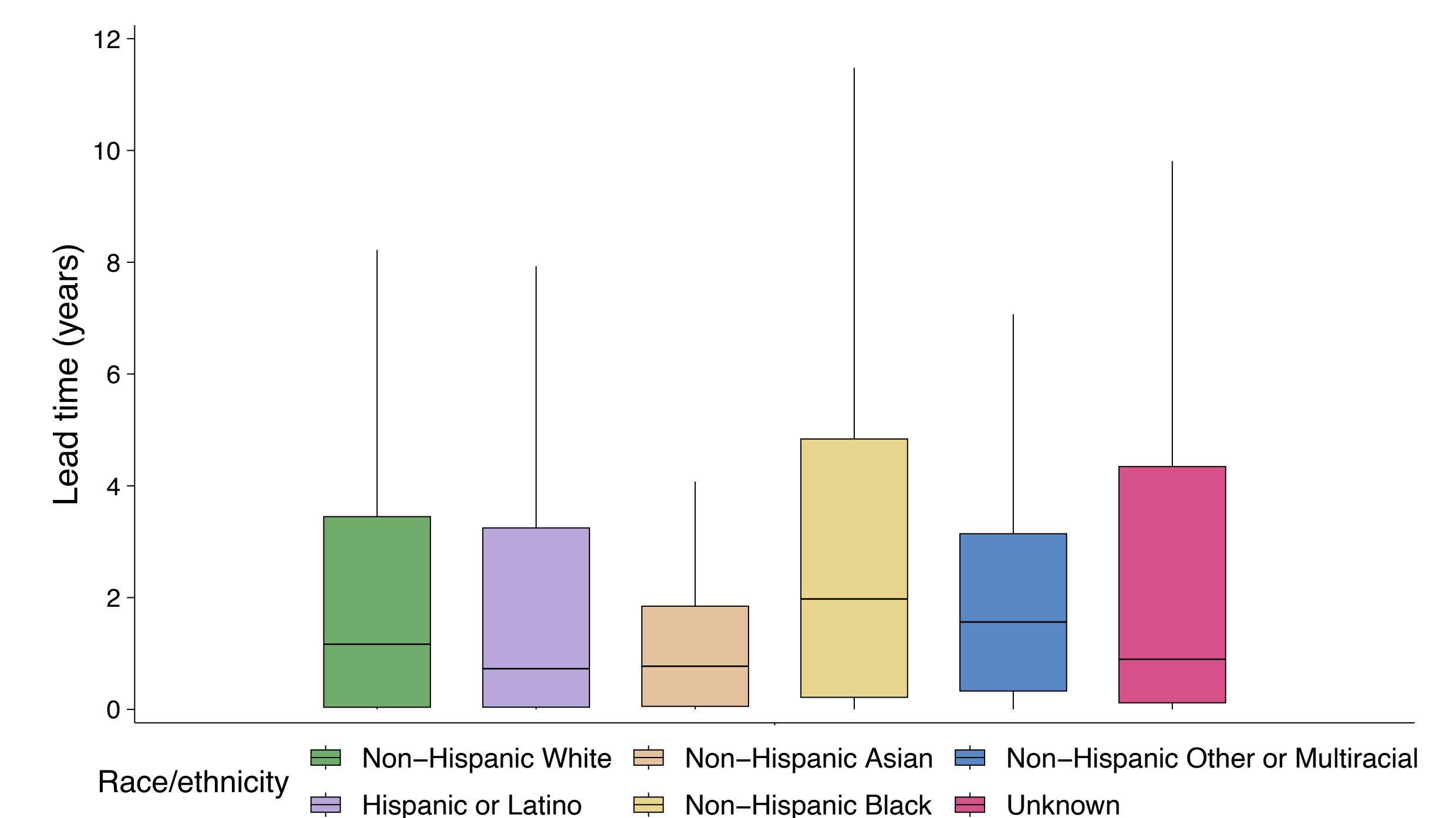
Rule	Sensitivity %	Specificity %	Cases Detected / Total Cases	Controls Correctly Excluded / Total Controls
BMI only	60.9%	63.8%	1028/1689	1242/1948
OR: BMI + hyperandrogenism	65.9%	63.3%	1113/1689	1233/1948
OR: BMI + hyperandrogenism + testosterone	70.2%	62.2%	1186/1689	1212/1948

Lead Time Subgroup Analysis

Lead Time Distribution Across All Race/Ethnicity



Lead Time Distribution by Race/Ethnicity



- Early detection precedes diagnosis by years: The framework identifies risk ~2.3 years earlier on average (median 1.2 years), with signals emerging across multiple visits prior to diagnosis.
- Lead time distributions varied by race and ethnicity, with the longest lead times observed in the Non-Hispanic Black group and the shortest in the Non-Hispanic Asian group, suggesting that this screening approach may offer the greatest potential benefit for earlier identification in the Non-Hispanic Black population.

Acknowledgments:

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